



Fox School of Business

TEMPLE UNIVERSITY®

Department of Risk, Actuarial, Healthcare Management, Legal Studies

AS 3504/5108

Actuarial Analytics

Analytics Practice Final Exam w/Key

Student Name: _____

Score: _____ / _____

This exam has 17 Questions for 120 Minutes. Complete all questions. Your lowest scoring Q will be dropped.

You need to show your work to justify all numerical calculations.



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1. Circle the true statements regarding statistical learning and briefly justify.
 - I. Model interpretability is positively correlated with model flexibility, meaning more interpretable models tend to be more flexible.
 - II. The proportional odds model is an example of a supervised, parametric model.
 - III. A classification model must include qualitative predictors.
2. Circle the true statements regarding model accuracy and briefly justify.
 - I. Deviance can be used as a measure of accuracy to assess generalized linear models.
 - II. Training data mean-squared error decreases monotonically as more parameters are added to a model.
 - III. A model's prediction accuracy is impacted by both irreducible error and reducible error.
3. Circle the true statements regarding the bias-variance trade-off and briefly justify.
 - I. As k in k -nearest neighbor increases, the variance will decrease.
 - II. If a model is adjusted to decrease variance, it usually comes at the cost of increasing bias.
 - III. If a model adjustment increases bias by a certain amount and decreases variance by the same amount, the expected test MSE will not be affected.



4. Circle the true statements regarding model validation and briefly justify.
 - I. Selecting models based on training error is a valid model validation strategy.
 - II. The holdout/validation set approach has the disadvantage of not being robust.
 - III. Leave-one-out-cross-validation will tend to produce more correlated test error estimates than k -fold cross-validation, where $k < n$.
5. Circle the true statements regarding ordinary least squares linear models and generalized linear models and briefly justify.
 - I. Multicollinearity represents a potential concern in linear regression models but not generalized linear models.
 - II. The hat matrix can help identify high leverage points in linear regression models.
 - III. A square root transformation can help address heteroscedasticity issues in linear regression models.
6. Circle the true statements regarding the linear exponential family of distributions and link functions and briefly justify.
 - I. The Student's t -distribution is a member of the linear exponential family.
 - II. The canonical link function for the exponential distribution is the log link.
 - III. The canonical link function for the normal distribution is the identity link.



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7. Circle the true statements regarding GLMs and K-nearest neighbors and briefly justify.

I. K-nearest neighbors differs from least squares regression because it is unsupervised.

II. If the true Bayes' decision boundary is very non-linear, a small K in K-nearest neighbors will perform best.

III. Deviance exactly equals residual sum of squares when the normal distribution is used.

8. A simple linear regression model uses # of times AI is mentioned in the prospectus to model its stock price at issuance. The correlation coefficient is found to be 0.6649.

# of times AI mentioned	Stock Price at Issuance
0	30
10	20
20	40
50	50
100	44

Calculate the linear regression equation and circle the statements that accurately reflect the parameter and interpretation.

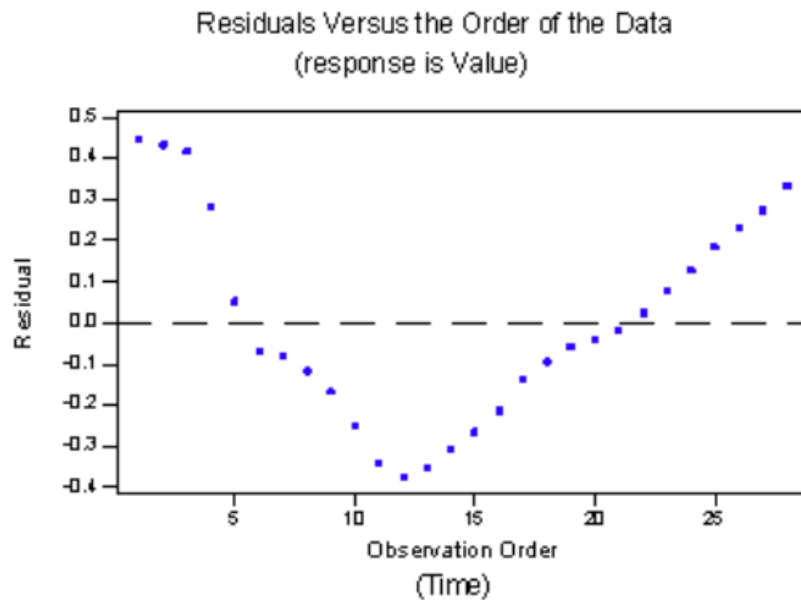
I. The predicted stock price when AI is not mentioned is predicted to be 29.75.

II. The predicted stock price increases by 0.25 on average every time AI is mentioned.

III. The proportion of variation in stock price that can be explained by AI mentions is 66.49%.



9. From the following graph of the residuals of a linear regression model, circle the statements that represent an accurate inference and briefly justify.



- I. A time series analysis may be necessary for this data.
- II. The model is missing a quadratic term.
- III. There is evidence of serial correlation in the data.



10. A dataset of 25 observations is fit to a linear regression model:

$$\hat{y}_i = b_0 + b_1x_{i1} + b_2x_{i2} + b_3x_{i3}$$

Data for the first three observations are provided.

The model mean squared error is 2.5267.

i	y_i	$\hat{y}_i$
1	1	0.6
2	2	2.3
3	3	3.0

$$h_{11} = 0.9, h_{22} = 0.7, h_{33} = 0.6$$

Calculate the Cook's distance for the second observation.

- A. Less than 0.04
- B. At least 0.04, but less than 0.05
- C. At least 0.05, but less than 0.06
- D. At least 0.06, but less than 0.07
- E. At least 0.07



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11. An actuary is considering one of the following three generalized linear models to model the odds of a claim, using income in thousands as a predictor.

Model I: Normal distribution with inverse link function

Model II: Logistic GLM

Model III: Poisson GLM with canonical link function

Using the most appropriate model and the parameters below, predict the odds of a claim for a policyholder with income of \$30K.

Parameter	Estimated Coefficient	Standard Error
Intercept	-4.8	1
Income	0.1	0.2

- A. 0.14
- B. 0.17
- C. 0.20
- D. 0.56
- E. 1.25



12. The following multiple linear regression model is used to predict a student's average hours of sleep in a week using two predictors and their interaction term.

$$\hat{y} = 7 - 2x_1 - 0.5x_2 - x_1x_2$$

Exam Week x_1 : 1 if Yes, 0 if No

Big Party x_2 : 1 if Yes, 0 if No

An actuary uses the same data and builds a new model, but codes No as 1 and Yes as 0.

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_1x_2$$

What is $b_1 + b_3$ for the new model?

- A. 0.5
- B. 2.0
- C. 2.5
- D. 4.5
- E. 5.0



13. A multiple linear regression model to predict movie revenue using four explanatory variables is fit to 25 observations. Release is a categorical variable with three levels. From the following summary statistics, find the number of explanatory variables that should be dropped from the model, using a significance level of 10%.

Beta	Point Estimate	Standard Error
b_0 : Intercept	20	15
b_1 : Budget	1.1	0.5
b_2 : # Actors	0.4	0.4
b_3 : Years in Production	3.1	1.8
b_4 : Release – Summer	25	8.0
b_5 : Release – Winter	-10	7.0

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4



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14. Given the data below on candidate generalized linear models, test the significance of the interaction term using the likelihood ratio test.

Variables	Maximized Log-Likelihood
Intercept	-900
Intercept, X_1 , X_2	-895
Intercept, X_1 , X_2 , X_1X_2	-893

- A. The test statistic is 2 and the null hypothesis is not rejected at 10%.
- B. The test statistic is 2 and the null hypothesis is rejected at 10%, but not at 5%.
- C. The test statistic is 2 and the null hypothesis is rejected at 5%, but not at 2.5%.
- D. The test statistic is 4 and the null hypothesis is not rejected at 5%.
- E. The test statistic is 4 and the null hypothesis is rejected at 5%, but not at 2.5%.



15. A multiple linear regression model is used to predict the ultimate losses using loss ratio x_1 and premium volume x_2 as two continuous predictors from six data points.

$$b_0 = 16, b_1 = 0.7, b_2 = 0.05$$

$$se(b_1) = s\sqrt{0.08}$$

The residuals are $\{-3, -3, -2, 1, 3, 4\}$.

Calculate the upper bound of the 90% confidence interval for the slope coefficient for loss ratio.

- A. Less than 1.5
- B. At least 1.5, but less than 2.0
- C. At least 2.0, but less than 2.5
- D. At least 2.5, but less than 3.0
- E. At least 3.0



16. A generalized linear model analyzes the number of likes on TikTok dance videos based on 3 categorical explanatory variables:

Length: 3 levels (Short, Medium, Long)

Time: 4 levels (Morning, Afternoon, Evening, Night)

Style: 2 levels (Old, New)

You perform likelihood ratio tests at the 5% significance level to compare Model I vs. II, II vs. III, III vs. IV, and IV vs. V.

Model	Predictors	Deviance
I	None	1,000
II	Length	990
III	Length + Time	970
IV	Length + Time + Style	967
V	Length + Time + Style + Length*Time	955

Of these 4 conclusions, how many of them are correct?

Model I is rejected in favor of Model II.

Model II is rejected in favor of Model III.

Model III is rejected in favor of Model IV.

Model IV is rejected in favor of Model V.

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4



17. A Tweedie GLM with $p = 1.65$, scale factor of 1, and log link function is used to predict accident severity.

Variable	Coefficient Estimate
Intercept	3
Location: Rural	-0.6
Time: Night	0.8

Calculate the variance of accident severity for a rural night event.

- A. Less than 25
- B. At least 25, but less than 50
- C. At least 50, but less than 100
- D. At least 100, but less than 200
- E. At least 200



Solutions:

1. Objective 1.a) Explain the types of modeling problems and methods, including supervised versus unsupervised learning and regression versus classification.
 - I. False
 - II. True
 - III. False
2. Objective 1.b) Explain the common methods of assessing model accuracy.
 - I. True
 - II. True
 - III. True
3. Objective 1.c) Understand how the bias-variance tradeoff impacts the selection of statistical learning methods.
 - I. True
 - II. True
 - III. False; $MSE = \text{squared bias} + \text{variance}$
4. Objective 1.d) Understand resampling methods used for model validation, including: Training set vs. test set approach, k-fold cross-validation, Leave-one-out cross-validation
 - I. False; overfit
 - II. True
 - III. True
5. Objective 2.a) Compare model assumptions for ordinary least squares and generalized linear models.
 - I. False; both
 - II. True
 - III. True
6. Objective 2.b) Identify the members of the exponential family of distributions and corresponding link functions.
 - I. False
 - II. False
 - III. True



7. Objective 2.g). Understand how approaches may differ compared to using an ordinary least squares model, including regularized regression (lasso, ridge regression), K-nearest neighbors
- I. False, both supervised
 - II. True
 - III. False, proportional to (differ by multiplicative constant)

8. Objective 2.c) Apply the business context of a problem to interpret parameters.
- I. True
 - II. False
 - III. False; R^2
- $$b_1 = 0.196$$
- $$b_0 = 29.75$$

9. Objective 2.d) Interpret diagnostic tests of model fit and assumption checking, using graphical methods
- I. True
 - II. False; there is serial correlation.
 - III. True

10. Objective 2.d) Interpret diagnostic tests of model fit and assumption checking, using quantitative methods
- $$e_2 = 2 - 2.3 = -0.3$$
- $$h_2 = 0.7$$
- $$D_2 = \frac{(-0.3)^2 * 0.7}{2.5267(3 + 1)(1 - 0.7)^2} = 0.069$$

11. Objective 2.e) Select an appropriate model, considering distributions and link functions
- Use Model II
- $$\ln\left(\frac{\mu}{1 - \mu}\right) = -4.8 + 0.1(30) = -1.8$$
- $$\hat{\mu} = \frac{1}{1 + e^{-(-1.8)}} = 0.142$$
- $$\text{Odds} = \frac{0.142}{1 - 0.142} = 0.165$$

12. Objective 2.e) Select an appropriate model, considering variable transformations and interactions
- $$\text{New } b_0 = 7 - 2 - 0.5 - 1 = 3.5$$
- $$b_1 = 7 - 0.5 - 3.5 = 3$$



$$b_2 = 7 - 2 - 3.5 = 1.5$$

$$b_3 = 7 - 3.5 - 3 - 1.5 = -1$$

$$b_1 + b_3 = 2$$

13. Objective 2.e) Select an appropriate model, considering t and F tests

Don't drop intercept b_0 (not explanatory)

$$df = 25 - 5 - 1 = 19$$

2-sided critical t at 10%: 1.7291

$$t_{b_1} = \frac{1.1}{0.5} = 2.2 \text{ (keep)}$$

$$t_{b_2} = \frac{0.4}{0.4} = 1 \text{ (drop)}$$

$$t_{b_3} = \frac{3.1}{1.8} = 1.722 \text{ (drop)}$$

$$t_{b_4} = \frac{25}{8} = 3.125 \text{ (keep)}$$

$$t_{b_5} = -\frac{10}{7} = -1.429 \text{ (would drop but since part of the categorical, keep)}$$

14. Objective 2.e) Select an appropriate model, considering the likelihood ratio test

$$LRT = 2(-893 - (-895)) = 4$$

$$q = 1 \text{ (difference in } df)$$

Critical value at 5%: 3.841, so reject at 5%, but not 2.5%.

Critical value at 2.5%: 5.024

15. Objective 2.f) Calculate and interpret predicted values, and confidence and prediction intervals.

$$SSE = (-3)^2 + \dots + 4^2 = 48$$

$$MSE = \frac{48}{6 - 2 - 1} = 16$$

$$se_{b_1} = \sqrt{16 * 0.08} = 1.131$$

2-sided critical t at 10% = 2.3524

$$0.7 + 2.3524(1.131) = 3.36$$

16. Objective 2.e) Select an appropriate model, considering the likelihood ratio test

Length uses 2 df , time 3 df , style 1 df , and the interaction uses $(3 - 1)(4 - 1) = 6$ df

$$\text{Model I vs. II: } LRT = 1,000 - 990 = 10$$

Critical $\chi^2_{2,0.05} = 5.991$, so reject H_0

$$\text{II vs. III: } LRT = 990 - 970 = 20$$

Critical $\chi^2_{3,0.05} = 7.815$, so reject H_0

$$\text{III vs. IV: } LRT = 970 - 967 = 3$$



Critical $\chi^2_{1,0.05} = 3.841$, so fail to reject H_0

IV vs. V: $LRT = 967 - 955 = 12$

Critical $\chi^2_{6,0.05} = 12.592$, so fail to reject H_0

First 2 conclusions are correct.

17. Objective 2.e) Select an appropriate model, considering distributions and link functions

$$\ln \hat{\mu} = 3 - 0.6 + 0.8 = 3.2$$

$$\hat{\mu} = e^{3.2} = 24.533$$

$$\text{Var}(y) = 24.533^{1.65} = 196.4$$